AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M109
.Net Programming

Final Examination
Spring Semester 2025/2026

Date: 13 / June / 2026

AOU EGYPT

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed: 2 Hours (120 Mintes)**

Instructions:

- Total Marks: **50 marks**
- This exam consists of **6 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

Part 1: Multiple Choice Questions (MCQ) [8 marks, 1 for each question]

1- What is the value of the item `finalArray[2,1]` after executing the following piece of code?

```
int[,] finalArray = {{17,21,33 }, {21,69,60 }, {33,77,44 }};
finalArray[2,1] = finalArray[2,2]++ + ++finalArray[0,1];
```

a- 67 b- 65
c- 60 d- 66

2- When one object is a specialized version of another object, there is an ……. relationship between them.

a- inheritance b- polymorphism
c- overloading d- overriding

3- When a base class member needs to be private to all outside code including derived classes and classes in the same package, you can declare that member as …….

a- private b- predicate
c- protected d- no given answer

4- In C#, we declare an output parameter by writing the ……. keyword before the parameter variable's data type.

a- result b- void
c- out d- return

5- When a method is ……., it means that multiple methods in the same class have the same name but use different types of parameters.

a- overridden b- overloaded
c- extension method d- concrete

6- Which statements are right regarding interface?

a. An interface specifies behavior for a class
b. It cannot be instantiated, but implemented by other classes
c. all methods and properties declared in an interface are implicitly `public`
d. All the above statements are right

7- The function ……. checks if `chr` is a (0 through 9) digit or not.

a- `char.Digit(chr)` **b- `char.IsPunctuation(chr)`**
c- `char.IsDigit(chr)` **d- `char.IsLetter(chr)`**

8- In some applications, some constructs are not intended to be instantiated to create objects, and they are designed for the purpose of providing an outline for subclasses. Such constructs are …….

a- concrete classes b- interfaces
c- extension method d- abstract classes

AOU EGYPT

Part 2: Short Questions (MCQ) [18 marks]

Question 2: In C#, both arrays and List<T> are used to store collections of data, but they have key differences primarily regarding size and flexibility.
Compare between Lists and array considering the following features: size, performance, and adding new elements. **(6 Marks)**

Question 3: Trace the following C# code segment and show its output. **(6 Marks)**

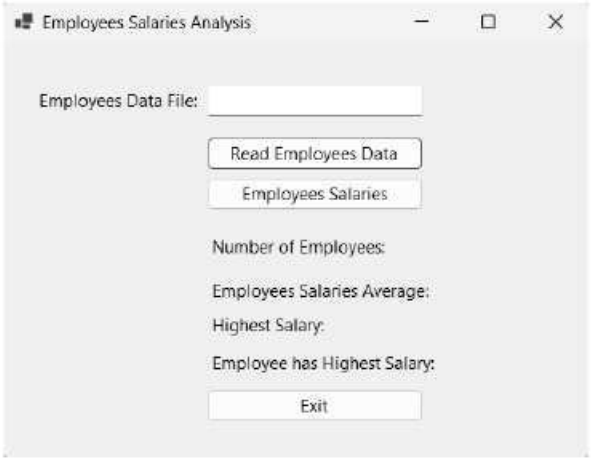
```
string senetence = ".Net Programming - Final Exam";
Console.WriteLine(senetence.Contains("final"));
Console.WriteLine(senetence.IndexOf(" - "));
Console.WriteLine(senetence.Remove(16));
Console.WriteLine(senetence.ToUpper());
var strBld = new StringBuilder(senetence);
strBld.Append(" Spring 25-26");
Console.WriteLine(strBld.ToString());
strBld.Replace(".Net Programming", "M109");
Console.WriteLine(strBld.ToString());
```

Question 4: Implement a C# function named **`GetLongestWord`** that accepts a List object that has a set of words and returns the longest word in the list. You need first to check if the list is empty or not. **(6 Marks)**

Part 3: Problem Solving Questions [24 marks]

Question 5: Write a C# program using Windows Forms that processes employees' salaries in a company as follows (consider the given form): **(12 Marks)**

- Read employees data (name, department, basic salary, bonus) (line per employee) from a file named **EmployeesData.txt**, where the salary value will be float and the bonus value is (0 to 1).
- Compute the average net salary of all the employees in the company as:
  $$\boldsymbol{net\ salary} = \boldsymbol{basic\ salary} \times (\mathbf{1} + \boldsymbol{bonus})$$
- Find the employee's name with the maximum salary in the company.



SOLUTION

Consider the above form that has the following points:

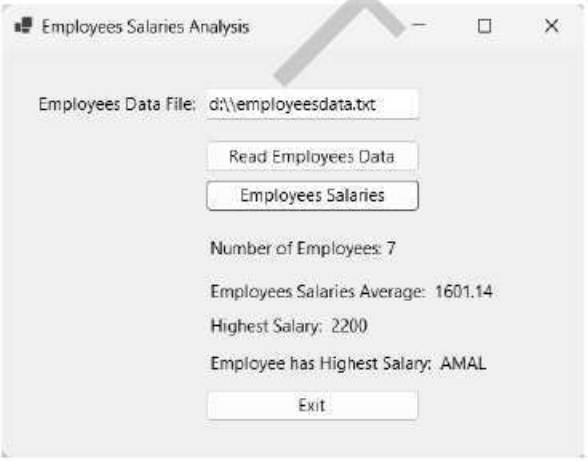
- A TextBox (named ***txtEmployeeSalaries***) to hold the file name.
- The form program includes a Button labeled "Read Employee Data" and named ***btnReadEmployeesData*** to read the employees data from the file and store them in 2 arrays, one for employees' names and the second to store the basic salaries, bonuses and the net salaries of employee. A Confirmation messageBox should be shown when reading is done successfully or the file does not exist. YOU CAN CONSIDER THE CLASS WILL HAVE 30 EMPLOEES AT MOST.
- The form program must include a Button labeled "Employees' Salaries Results" and named ***btn EmployeesSalariesResults*** to perform the math calculations as given above and the results will be shown in the following labels.
  - Label titled "Number of Employees" and named ***lblNoOfEmployees*** to hold the number of employees.
  - Label titled "Employees Salary Average" and named ***lblEmployeesSalariesAverage*** to hold the average of salary of employees ROUNDED TO 2 DECIMAL PLACES.
  - Label titled "Highest Employee Salary" and named ***lblHighestSalary*** to hold the highest salary of all the employees.
  - Label titled "Highest Salary Employee Name" and named ***lblHighestSalaryEmployeeName*** to hold the name of the employee with the highest salary. THE NAME MUST BE IN CAPITAL.
- Provide a MessageBox notifying the report is ready or the input file is missing.
- The form program must include a Button labeled "Exit" and named ***btnExit*** to close the program when press it.

Write the code of the 3 buttons and the necessary data field declarations and initialization.

Sample input file:

```
Ahmed 1000 0.07
Amal 2000 0.10
Nora 1500 0.14
Ayman 1550 0.12
Samir 800 0.25
Joudy 1200 0.16
Aziz 1750 0.20
```

Corresponding program output:



YOUTIGYPT

Question 6: **(12 Marks)**

A. Write the **Account** class (to treat a bank account) which includes: (follow OOP principles studied in the course) **(4 Marks)**

1. Three fields: **Owner**, **Number** and **Balance** to store the **Account** owner name (e.g. "Ahmad"), account number (e.g. "AB103456") and balance amount (e.g. 1576.25).
2. Getters and setters for the attribute three fields.
3. A Parameter constructor with the three fields.
4. A Method **Deposit(double amount)** that credits a given amount to the balance. The amount to be credited and the new balance should be printed.
5. A Method **Withdraw(double amount)** that debits a given amount from the balance. The amount to be debited and the new balance should be printed. If the to-be-debited amount is greater than the balance, a meaningful message should be printed.

B. Write the **SavingAccount** class as a child class of the **Account** class which includes: **(4 Marks)**

1. A Fields **InterestRate** to store the accounts annual interest rate percent (e.g. 4.5).
2. A Parameter constructor with 4 parameters including the inherited ones from the base class.
3. A Method **ApplyInterest()** to update the balance according to the interest rate value. **The new balance value = balance x (1 + interest rate).** The new balance should be printed.

C. Write the Main method to implement the following: **(4 Marks)**

1. Create a List reference with 2 SavingAcoount objects with some values.
2. Apply the IntrestRate for the 2 objects in the list.
3. Deposit 500$ to the first SavingAccount object in the list.
4. Debit 350$ to the second SavingAccount object in the list.

End of questions

AOU EXAM